

[00:02] All right.

[00:02] So this is our lecture week eight, and in this lecture we're going to talk about, um, analyzing time series data or, uh, time series data analysis.

[00:17] We're going to talk about generating, uh, date ranges at different time intervals, uh, re-indexing, uh, resampling, uh, that includes downsampling and upsampling, a bit more emphasis on downsampling.

[00:33] Uh, we also learn about creating and plotting rolling windows, um, explaining how re-indexing time series data can improve plots, and, uh, that sometimes it can introduce, uh, missing values and how we can fix that.

[00:50] Uh, we need to be able to describe, um, how to apply a function, uh, to, to, to, to fix missing values in data that's indexed by dates.

[01:03] Also, um, we need to-- uh, we should be able to distinguish between downsampling and upsampling, uh, explain the use of the label and close pa-- uh, cl- label and close parameters in resample method that is used for resampling.

[01:24] And, uh, we also need to be able to explain how downsampling can improve plots.

[01:32] Uh, we should also be able to describe the use of, uh, rolling windows, uh, by the end of, uh, this lecture.

[01:46] So before digging into these, um, aspects and covering these objectives, uh, we first want to know what is time series data.

[01:57] What do we mean by that?

[02:01] All right.

[02:01] So time series data.

[02:04] Uh, time series data consists of a series of observations that are recorded over a period of time.

[02:10] Each observation is associated with a specific, um, uh, time point.

[02:16] And of course, time series data can be collected from different application domains such as finance, economic, health, engineering, uh, energy and so on, as we'll, we'll, we'll see, uh, in the coming slides.

[02:33] One of the application domains of time series is, uh, finance.

[02:40] Uh, it is used to predict market trends, estimate stock prices, and identify, uh, trading opportunities.

[02:50] Uh, you can see a couple of examples of, uh, time series data that's been plotted or been collected and plotted from, from market, from markets, from stock market.

[03:01] And, um, so here you can see the data that's collected from stock market.

[03:08] And on the right-hand side, you can see the, um, the, the ratio of, uh, the, the currency exchange rate, uh, from, uh, Australian dollar to US dollar.

[03:22] You can see the trend over a period of, uh, one year.

[03:26] And, um, this is taken from Google.

[03:29] You can just play around and see the trend across, uh, over, over different time intervals.

[03:35] Uh, so you can actually, uh, focus on a specific time interval if you are more interested in that time period and see how the, the, the, uh, exchange rate has changed.

[03:47] Uh, same thing, uh, you can do the same thing for, uh, for stock price, uh, um, uh, for, uh, for different time intervals.

[04:00] Another application domain is in healthcare.

[04:02] This graph is taken from, um, uh, health.gov.

[04:10] Uh, it's from Australia.

[04:12] You can see that, uh, the death rates associated with COVID-19 has been, uh, plotted in this, in this graph from January, um, uh, twenty-two to March twenty-three.

[04:30] That's the time period.

[04:31] And, uh, you can see the-- you can see how deaths have changed over time.

[04:38] Uh, in healthcare, time series data is used to monitor vital signals of patients or track diseases, outbreaks.

[04:47] In this case, one of the examples of outbreak, of course, is COVID-19.

[04:52] And analyze medical records to identify patterns and predict treatment outcomes.

[04:58] Uh, for example, you can have a graph or have a plot that shows how people have, um, uh, or have, have used vaccination, or how they have got, got vaccinated, and maybe compare that against the death rates and see if

[05:15] there, there's any connection.

[05:16] So you can, uh, you can compare, um, different datasets, uh, or different, uh, time series, um, time series data, uh, across, uh, um, a period, let's say one year or two years or the whole period of the, uh, COVID-19, um,

[05:36] impact, and then, um, maybe find some insights by m-making those comparisons.

[05:45] Uh, another application domain is energy.

[05:52] Um, you might be interested to monitor energy consumption patterns or predict demand, optimize production schedules, and monitor equipment performance.

[06:03] Um, so All these are examples of using time series data in, in, uh, in energy domain.

[06:11] And you can see an example of that here, the petrol prices in Melbourne from, um, nineteenth of March to, uh, thirtieth of April, and how it has changed, um, over this, this time interval.

[06:30] You can also use time series data in a co-- in, um, uh, weather forecasting.

[06:36] It can be used to, um, to track weather patterns, uh, monitor climate changes, and predict natural disasters such as, uh, floods, for example.

[06:51] Here's an example of, uh, time series data that shows, uh, the temperature of, uh, Melbourne Olympic Park, um, uh, from Thursday to today.

[07:04] You can see how the temperature has changed.

[07:10] So we can look at the patterns and identify the trends, and then use those trends for making, uh, predictions about, for example, what would be the next, uh, value for the temperature, or what would be the next value for the

[07:24] stock price, or what would be the next exchange rate for the currency, and so on and so forth.

[07:33] Um, another application domain for time series data is to monitor social media data or to identify patterns or trends in, uh, user behavior in social media.

[07:47] Uh, you have-- you can see two examples here.

[07:51] Um, one shows, uh, the one at the bottom, um, shows the, um, the prevalence of different, uh, or the mean emotion score for, uh, anger, sadness, optimism, and joy, um, that's been extracted from user replies to COVID-19 news.

[08:13] So this is, um, uh, this graph-- plot has been discussed in a publication that, um, um, that you can, you can find, uh, by following the link that's been provided, and you can see more discussion about that.

[08:28] But you can see, for example, the anger has been, um, based on this graph, um, uh, m-mean emotion score for an-anger has been higher than, uh, sadness, optimism, and joy.

[08:42] Um, so you can actually extract user comments or user replies, uh, or tweet contents from different entities or different users on social media or on Twitter, for example, or, uh, or YouTube or Instagram, different, different social media platforms, and analyze,

[09:04] uh, analyze the data using techniques such as sentiment analysis and emotion analysis, NLP techniques essentially allows you to do that.

[09:16] And then you can, um, you can plot your observations over the time interval over the period of the study, whether it's one year, two year, whatever it is.

[09:25] And then you can find some perhaps trends and patterns that could suggest, um, uh, or could, could be used to hypothesize certain, uh, hypothesis and, uh, that could be further investigated and so on.

[09:45] Um, we also have, uh, provided a platform, uh, from Covaxi, uh, tool that you can, uh, try to browse yourself, and you can see some interactive plotting there, where you can see the, uh, uh, see some data about COVID-19 vaccination

[10:04] that's been plotted.

[10:06] And you can, you can interact with the plot and see the, see the data during different time intervals and so on and so forth.

[10:12] You can, you can, um...

[10:14] I think, uh, it allows you to also add your own, um, keywords to filter the results and so on and so forth.

[10:20] So it's an interesting place that we can explore some, uh, time series data.

[10:28] Uh, there are also other applications in transportation, in manufacturing, and IoT or Internet of Things, uh, which you can explore, and you can look at more examples once you know more about time series.

[10:49] I've also provided a video for you here.

[10:51] You can watch this video, uh, about time series and some of the applications in real world.

[10:56] Uh, I'm gonna skip it for now, but, uh, uh, uh, strongly recommend watching the video.

[11:07] Um...

[11:14] Okay, how to reindex time series data.

[11:25] So one of the things that we can do, uh, or apply to our, um, time series data is to reindex it.

[11:35] Reindexing is a powerful technique for manipulating and transforming time series data.

[11:40] So we might-- what is the purpose of having time series data?

[11:44] We want to analyze the data, right?

[11:46] We want to find some interesting observations and then report them or use them for, um, you know, drawing some hypothesis or making predictions.

[11:56] Uh, so to be able to do that, we need to do-- we need to, um, analyze the time series data and to, to make a-- to, to do the analysis, we, we might need to do reindexing.

[12:08] Reindexing helps us to And manipulate and transform time series data.

[12:13] Um, applications of reindexing includes the resampling.

[12:17] So it helps us to resample the data at different frequencies.

[12:22] Uh, we'll, we'll talk about resampling in a bit.

[12:24] Uh, aligning, which allows for, um, um, aligning multiple time series data or datasets that have different frequencies, uh, and that makes it easier to compare those, uh, those datasets.

[12:40] Eh, reindexing is also used in shifting, where, uh, we can shift the time, uh, the, the time index by a certain offset, uh, again, to make it easier to, to compare, uh, different datasets and to align data from, from different

[12:56] sources.

[13:03] Uh, we use function `date_range` for reindexing.

[13:09] Um, what `date_range` returns is a date time index object.

[13:16] So, uh, we call `date_range` function.

[13:20] We pass the start date, the end date, and the frequency as the parameters.

[13:25] So `start_date`, `end_date`, and frequency.

[13:28] And the function returns a date time index for us.

[13:35] The frequency can come in, uh, um...

[13:38] It can be provided or can be specified as business days, uh, calendar day, weekly, monthly, semi-monthly, which is every, uh, fifteenth day or, uh, the f- uh, half a month, uh, almost half a month, and, uh, so on and so

[13:56] forth.

[13:56] So there are different values that we can provide to specify different, different frequencies.

[14:04] Here's an example of, uh, generating time periods using, um, um, for example, MS here.

[14:14] So f-f- we have set, uh, we have set the frequency to MS, which means, uh, here we create a monthly date that range from-- ranges from the start date to the end date with the frequency of month start.

[14:30] So the, the frequency is monthly, but we are specifying that by, by MS.

[14:36] We are, we are saying that we want the start of the month.

[14:39] So the data that is going to be, um...

[14:43] So, so the range that is the, the, that is going to be generated for us inclu- w-would be, would be monthly, but, uh, only this, uh...

[14:52] but the value that is, is selected for, uh, for the day of the month would be, uh, the start of the month.

[14:59] In other words, we create a range of, um, days with frequency of monthly.

[15:06] So the first, uh...

[15:08] and, and, and, and the data points would be the-- the data point that we pick for each month would be the first day of the month.

[15:15] So we'll have the first day of the first month of twenty-twenty, the first day of the second month of twenty-twenty, the first day of the third month of twenty-twenty, and so on and so forth.

[15:25] So we are picking the first days of every month in twenty-twenty, which is the time period, uh, the first month of twenty-twenty, uh, to the last month of twenty-twenty.

[15:37] And so this is the range that is generated by this function, by `date_range`.

[15:43] Okay?

[15:43] It's a date time index that we can use for, for example, resampling, as we'll see.

[15:51] Um, here's another example where we specify the frequency to be business days.

[15:56] So the frequency is set to B.

[15:58] B, uh, stands for business days.

[16:00] And, uh, in this case, we are selecting, uh, th-this, uh, sorry, this date_range function returns a range of dates, uh, that start from the first month of twenty-twenty and, uh, goes all the way up to, um...

[16:17] Yeah.

[16:17] So, so th-this is the first month of twenty-twenty.

[16:19] Uh, it starts from the first day of the first month of twenty-twenty, goes up to the, uh, uh...

[16:24] goes up to, um, the la- uh, last day of the first month of twenty-twenty.

[16:30] So basically, um, January of twenty-twenty from the first day to the last day.

[16:36] But the frequency is business day, which means it's going to exclude some of the days.

[16:41] It's going to exclude, um, uh, weekends, right?

[16:46] So what we're getting here is that, uh, first, first of January, se- second of January, third of January, and then f- we don't have fourth and fifth of January here 'cause, um, they have been, uh, weekends, Sa-Sa-Saturday, Sunday.

[17:01] And then we have sixth of January, and then it continues to seventh, eighth, ninth, tenth.

[17:07] And then again, we don't have eleventh and twelfth 'cause they have been weekends.

[17:11] And then start-- we, and then we continue with, uh, thirteen and so on and so forth.

[17:16] So, um, this, this, um, date_range function in this case, uh, generates only business days because we have set the frequency to be business days.

[17:31] Here's another example.

[17:32] We-- This time we specify the frequency to be the f- the Mondays of the week.

[17:38] So we have weekly frequency.

[17:42] Uh, we have weekly frequency, but, um, the day that we are going to pick or the day that is going to be generated will be the Monday, uh, will be the Monday of that week, right?

[17:55] So in other words, the frequency is every Monday, right?

[18:00] Every Monday.

[18:03] Um, so what was the frequency here?

[18:05] Ev-every business day.

[18:06] Here is every Monday.

[18:08] So every Monday of the week, we have, um- Uh, so we have, uh, seventh, and then we have fourteenth, and then we have twenty-first, and so on.

[18:19] So we have four Mondays in, uh, January of, uh, twenty twenty.

[18:25] Again, we have start date, end date, and then we're picking every Monday from this range.

[18:30] So we have a new date range that is created by this new frequency, which is, uh, a weekly frequency, uh, where we're picking the Mondays.

[18:41] In other words, by every Monday frequency.

[18:46] Okay.

[18:49] Okay, so here we have another example.

[18:51] We have, um, used date range function with frequency of twelve hours between the range of, uh, first of January twenty twenty-two and, uh, end of January twenty twenty-two.

[19:04] Uh, this is going to print, as you can see here, um, uh, print, um, the, the date times every twelve hours starting from midnight of January first, um, till midnight of, uh, uh, the end of January.

[19:25] Um, uh, the hour twelve of thirty-first of January is not included because, uh, the date range function is, uh, right exclusive, so it does not include the, uh, the last, um, date time.

[19:47] All right, uh, so here we have an example of reindexing with, uh, date time indexes.

[19:55] Uh, we first create a data frame from stock data that's stored in this file, CSV file, and, uh, we're extracting these particular columns.

[20:04] So we're not interested in all the, all the columns that are stored in this CSV file.

[20:09] We're only ex-extracting the date, open price, high price, low price, and close price.

[20:16] Uh, and we're also passing date-- We're assigning date as, uh, to, to, to parse date's parameter of read_csv function, which essentially tells the CSV function-- the read_csv function to, uh, treat date as a date time object.

[20:34] Uh, so here we have, uh, displayed the first three rows of, uh, of the dataset, uh, or the data frame.

[20:44] Uh, but of course, we know that there's more data here that corresponds to, uh, to different, uh, different month-- different days, uh, from different months of, uh, uh, twenty twenty.

[20:59] And, uh, here, here we create a range that starts from, uh, January of, uh-- the first, first of January of twenty twenty till, uh, end of, uh, December, uh, of twenty twenty.

[21:15] So the whole, uh, twenty twenty is covered.

[21:18] And the frequency of the dates, uh, or the frequency of the, uh, time points that are generated is specified here.

[21:28] Uh, that is, uh, every Friday.

[21:32] So it's weekly and Friday of the week.

[21:36] So it means every Friday.

[21:39] Uh, so we're extracting from, uh, the next step is that we, we reindex the dataset, data frame using this new, uh, new index, Fridays.

[21:49] Uh, and that results in, uh, extracting all the data points from the original dataset that are from Fridays.

[21:58] And we, we exclude all other data points from the data frame by, by, by, by reindexing based on Fridays.

[22:05] So stock data that reindex Fridays is going to give us, uh, the data that corresponds with Fridays.

[22:12] Uh, which means, uh, as you can see, uh, third of January, tenth of January, seventeenth of January, and so on and so forth.

[22:20] Here we have only displayed the first three, uh, Fridays, uh, correspond to the first, you know, to, to the first three Fridays of January.

[22:29] Uh, here also you can see that the data that correspond to these Fridays have been extracted from the original dataset.

[22:37] And, uh, that includes this, this row here that corresponds to, um, the first Friday of January.

[22:47] And, uh, yeah.

[22:53] Okay.

[22:53] So, uh, in the previous example, we saw that, um, only the rows that correspond to Fridays will remain in the reindexed, uh, data frame.

[23:05] But the problem is, um, some of these Fridays might be public holidays, um, in twenty twenty.

[23:14] Uh, it turns out that Good Friday and, uh, Independent-- Independence Day and, uh, Christmas, these three, these three days are Fridays.

[23:24] And, uh, there will not be any stock data for these, uh, for these days, which leads to missing values in the resulting data frame.

[23:34] That means in the, in the reindexed data frame.

[23:37] So reindexing data, da-data frame with, uh, or, or reindexing the original data frame with a date time indexes that we generated, uh, which is, uh, which was, uh, the Fridays of the, uh, the Fridays of the weeks, um, uh, has

[23:54] led to, uh, some, some, some missing values.

[24:00] Um, so here we have another example that we reindex with a semi-month index.

[24:06] This time, we divide every month into two parts, and, uh, that means the, the date range function divides every month into two parts and picks the beginning and the middle of the month.

[24:19] Um.

[24:20] And creates a range, uh, accordingly.

[24:22] So you can see that, uh, here we have printed out the first four items in the semi-month, uh, uh, what do you call it, uh, date time index.

[24:33] And you can see, uh, it contains the first day of January, fifteenth of January, first day of February, fifteenth of February, and so on and so forth.

[24:40] So this is, uh, this only the first four, four items in the date time index, which is semi-month.

[24:47] Uh, now it's time to use semi-month as the index for re-indexing.

[24:52] So, uh, again, uh, we, we can call the, the, the re-index function, um, to, to, uh, to re-index based on semi-month.

[25:03] So re-index using semi-month, uh, it will give us the re-indexed, uh, uh, stock data and, um, and only p- this will obviously only pick, uh, the, the dates that are in the range, uh, which will lead to, um, uh, the

[25:26] missing value issue again that we'll discuss in the next slide.

[25:33] All right.

[25:33] So the missing value issue in this example is that, um, of the two dates, uh, that means the first and the fifteenth of each month, uh, these dates, uh, often fall into, uh, or land on, uh, weekends.

[25:54] And, uh, since there's no trade over the weekends, um, in the stock market, we won't have any data for those dates, uh, which leads to all these missing values that you can see here.

[26:07] We have NaN for this date, and this date, and this date, uh, which is not what we desire.

[26:15] And when we plot this data, this-- then we get, uh, something like this.

[26:20] So these are all the missing values here, here, and here.

[26:26] All right.

[26:27] So to fix this missing values issue in the example that we just explored, we can define a custom function or a user-defined function.

[26:36] In this case, we're defining the `adjust_date` function that takes a date as its input.

[26:41] If the date-- if the weekday of the date is less than five, that means the date is M- Monday to Friday, then that's fine.

[26:50] We don't make any changes and just return the date as it is.

[26:53] But if the da- the weekday of the date is five, uh, we're starting from zero, so zero, one, two, three, four, five.

[26:59] Five would be Saturday.

[27:00] If it is Saturday, then, um, we change this to da- di- to the, to the day before Saturday, which is the previous Friday.

[27:09] So date minus dt.timedelta days equals one, uh, does that for us.

[27:16] So this actually, this part, dt.timedelta, days equals one, will create one day for us and date minus one will, uh, will change-- will, will, uh, will go back one day, in this case, to Friday.

[27:33] And, uh, that date will be re- date will be returned.

[27:38] Uh, however, if weekday is six, which is the other, uh, scenario, uh, if weekday, weekday is six, that means it's, it's a Sunday.

[27:49] In that case, uh, instead of, uh, going back, we go forward.

[27:56] So we actually move to the next Monday.

[27:59] In other words, uh, when we don't have data, that means when the date is a Saturday, then we use the da- the data that comes from its, uh, previous Friday.

[28:12] Um, and to be able to do that, we change the date here.

[28:16] And if the date falls into, uh, if the day, um, uh, what do you call it?

[28:25] Uh, becomes a Sunday, then instead of the data from Sunday, which doesn't exist, we use the data from Monday.

[28:36] Um, for that, we, we move the date one day forward in this case.

[28:46] So, um, you can see in this, uh, this slide that we have a-- we applied the adjust_date function to, um, to the, um, to the range that we created.

[29:02] So we created a semi-month range, and then we convert that to, to a series, and we apply the adjust_date to all members of the series, which are the dates.

[29:13] And if, uh, there are any dates that are Saturdays and Sundays, they will be adjusted accordingly, uh, based on the adjust_date function.

[29:20] And, uh, then we get a, um, a, a semi-month adjusted, um, uh, uh, range.

[29:33] So we use this adjusted range or adjusted semi-month, semi-months, um, uh, to, to, to do the re-indexing for the stock data.

[29:41] So stock data re-index, and this time instead of semi-month, we use, uh, semi-months adjusted, uh, with the adjusted dates.

[29:50] As a result, uh, uh, when we plot the data, we get with the graph, we get a graph like this, which, uh, doesn't contain the missing values, or in other words, we have fixed the these missing values with, uh, with,

[30:03] with this adjustment that we've made to the dates.

[30:07] All right.

[30:08] Uh, you can see still we have one row here that contains, uh, missing values, uh, which could be removed from the data, from the, uh, data frame for visualization.

[30:19] But the reason we have missing values is the, there is that we don't have a date that comes Uh, before that.

[30:26] So it's...

[30:26] If this is a Saturday, uh, of course, we can look at the calendar to see what day exactly it is.

[30:31] But if it's a Saturday, uh, we don't have a day, we don't have a Friday before that, clearly, so it cannot be adjusted, so values remain the same.

[30:38] If it's a Sunday, um, maybe there hasn't been any, um, uh, value available for its, uh, subsequent Monday, and that's why the values are nil.

[30:49] So we can look at the calendar to see actually which, which, which, which one has been the case.

[30:54] Um, but, uh, we don't have values for, for the first row.

[31:04] Okay, how to resample time series data.

[31:10] But what is resampling?

[31:12] Resampling is changing the frequency of, uh, uh, the time series data.

[31:17] So we might want to reduce the frequency or increase the frequency depending on the application and what we need to do.

[31:24] This is useful for adjusting the data to match the frequency of another data set, reducing noise, preparing the data for forecasting and computational storage efficiency.

[31:35] So these are different reasons we might want to resample our, uh, data set.

[31:42] There are two main types of resampling as I mentioned.

[31:45] Uh, downsampling, which means we reduce the frequency of the data, data points or the frequent-- uh, uh, the frequency, uh, of the, uh, uh, of the s- time series data, or we upsample, which means we increase the frequency of the

[32:02] time series data.

[32:06] So, um, in the case of downsampling, we might want to reduce the frequency of the data by aggregating or summarizing it over, uh, larger time intervals.

[32:20] And, uh, this happens when, when you have, uh, high frequency data, but you only need to analyze it, uh, at a lower frequency.

[32:30] Uh, also when you need to reduce the size of data because of the storage or computational efficiency.

[32:36] Sometimes you don't need all the data that's been collected.

[32:39] The frequency at which you're receiving the data or you're collecting the data is not necessary for your analysis.

[32:45] So, um, it makes sense to reduce that frequency for better, uh, efficiency, computational storage efficiency.

[32:54] Uh, also it will help reduce noise.

[32:58] Um...

[33:02] Okay, so for instance, the-- a minute-level stock price or stock price data can be downsampled, uh, into, uh, daily summaries.

[33:12] So instead of having the data for stock price every minute, we will, we'll downsample that.

[33:19] We already have the data, let's say, for, for every minute, but we don't need, uh, uh, you know, we don't need that because maybe there's not, uh, there's not much, uh, change, you know, in every minute.

[33:31] So we may want to downsample that to actually increase the, um, uh...

[33:37] Decrease the frequency of the, uh, um, time series data to, for example, daily.

[33:46] All right, so to downsample, we can use function resample.

[33:50] Um, here we are using, uh, um, acres burned.PKL file to create a data frame, acres burned, and then we use resample function.

[34:05] We specify the rule pro-- We, we set the value of the rule par-parameter of the sample function to M, um, which, uh, essentially specifies that we are going to, um, to downsample, uh, to, to, to the frequency of month.

[34:23] That means, mm, uh, to monthly res-resampling or, or monthly downsampling.

[34:33] Uh, so what happens here is that, uh, of course, we have the data for all the dates, and the data will be summarized using the sum function.

[34:43] The data will be aggregated for every month using the sum function.

[34:46] So we calculate the sum of the acres burned for every month, and, uh, that will be displayed in the...

[34:53] Or that will be captured in the, uh, in the resample data frame.

[34:57] So in the resample data frame, you can see that we have only one data point for every month.

[35:01] So this is for, uh, for the, for, uh, the first month.

[35:05] That is for January, February, March, and, uh, the end date of each month is used here to, to label each month.

[35:14] And, um, the, the value that is here is the summation of the, the acres burned, uh, in that, in that particular month.

[35:24] So this is the value for January, this is the value for February, and this is the value for March in, uh, 1992.

[35:40] Here we have another example of downsampling, but this time we use two parameters, label and closed.

[35:48] Um, we are setting the parameter label here to right and close to right, which means, um, when we set the label right, uh, the label to right, it means the labels for the resample data should be the end of each

[36:04] quarter.

[36:05] That means the last day of each-- of the quarter.

[36:09] And when we set the closed to right, it means that each interval includes the end, the end point, or, uh, the last data point or the last day of the interval.

[36:23] And the rule here is set to Q.

[36:25] That means, uh, the intervals will be quarters So we divide one year, the date of re-- the, the, the range, uh, that we're, we're using is, is, is a, is a whole year.

[36:35] So we are dividing the year into, uh, four quarters.

[36:39] That means each quarter will be three, three months, right?

[36:43] And, uh, because the label is set to right, uh, the labels, uh, for the resample data should be, uh, the end of each quarter.

[36:53] Because when we consider a quarter, a quarter has a beginning and an end, right?

[36:58] So when we use a label here, the label should-- we need to decide whether we want it to be the beginning of the quarter or the end of the quarter.

[37:06] By setting label to right, m-it means that we're using the, um, uh, the date at the, at the right-hand side of the-- or at the end of the quarter, which means the last day of the quarter to be the, to

[37:20] be the label in the new data frame.

[37:23] And, uh, and a-again, when we set the close to right, that means we, we include the, we include the value of the, uh, acres burned that corresponds to the last day, uh, into our, into the calculation of sum, uh, as

[37:39] well.

[37:41] Right?

[37:53] Okay, so here's an-an-another example.

[37:56] In, uh, in this example, we are setting the label to be left, which means we are going to this time choose or pick the beginning of the quarter as the label, right?

[38:09] Uh, so label left means we are, we are picking the beginning of the quarter.

[38:14] And close is set to left, which means, uh, the data point from the left, that means the beginning of the quarter, uh, will be used in calculating the sum for each interval or for each quarter.

[38:28] Here, our int-- the length of our interval is, or the width of our interval is a quarter of the year.

[38:33] It's three months.

[38:36] So, uh, yeah.

[38:40] So, uh, this is what we get, and you can actually compare, compare this against what we had in the previous example.

[38:48] You can see that for the first row, we had the value at the end of the quarter, but here we have the value at the beginning of the quarter, which goes back to nineteen ninety-one.

[38:59] Here we-- Here, here, the end of the quarter is nine- in nineteen ninety-two, um, at the end...

[39:06] at the, at the last day of March.

[39:08] Uh, but here you can see that the beginning of the quarter is used, and the value is, uh, the, what do you call it?

[39:18] Uh, the value is, uh, the value that is used for the label is the beginning of the, uh, quarter, that is, uh, the end of December nineteen ninety-one.

[39:30] And so on and so forth.

[39:30] So you, so you can compare the values.

[39:36] Okay, here's a question for you.

[39:38] Um, what happens if you set the label to left and close to right?

[39:44] Uh, try executing this and see what you get, and compare against, uh, the data frames that we just created.

[39:54] Uh, this question is for you to further practice.

[40:00] All right.

[40:00] We can use downsampling actually to improve plots.

[40:04] How?

[40:06] When plotting time series data with a large number of data points, it can become difficult to see the overall trend.

[40:13] Uh, when we have too many fluctuations, then, then the data becomes noisy because we have too, uh, too much, uh, variability.

[40:22] Then downsampling can help by reducing the number of data points that are displayed on the graph, therefore, uh, smoothing out the, the, the graph or the plot and making it easier to identify the, the patterns or the trends.

[40:38] Um, this is particularly helpful when visualizing long-term trends, uh, or when, uh, we want to, uh, present our data, uh, which makes it easier for people to see the, see the patterns or see the trends.

[40:53] When you want to, for example, create a report from the data.

[40:57] Uh, and yeah.

[41:02] So example is when-- if the original data time series, uh, if the original time series data includes daily stock prices, uh, we can downsample it to weekly, uh, or, or monthly da-monthly data points, which can help, as I said, uh,

[41:17] to smooth out the plot and make it, and make it easier to identify if there's any patterns in the trades or the, uh, to find the patterns or the trends in the trades in the, in the stock market.

[41:33] All right.

[41:34] Um, downsampling.

[41:38] Um, yeah.

[41:38] So here's an example of improving, uh, plots using downsampling.

[41:43] Uh, this is the original plot.

[41:47] You can see that there are many fluctuations.

[41:51] The, uh, variability is high.

[41:53] So that might suggest to, uh, improve the, uh, readability of the plot or visibility of the plot by downsampling.

[42:06] So here's an-- here's the downsampled, uh, the visualized downsampled data or plotted downsampled data.

[42:15] Uh, we are resampling the data.

[42:19] This is downsampling because how do we know it is downsampling?

[42:22] Because the original-- the frequency of the original data set was higher, was daily, and now we're reducing that to weekly.

[42:29] So rule equals week, that means we are sampling the original data set Um, uh, weekly.

[42:37] Okay?

[42:38] And, uh, we are using mean to aggregate the values.

[42:43] So we are considering the, the weekly average of, um, the close price, right?

[42:52] So for every day, we consider the close price, and we compute the average of those close prices for, for the whole week.

[43:01] And, uh, that will give us the value for that particular week, right?

[43:06] And then that will reduce the frequency of the resulting data frame and therefore makes our, uh, our plots, uh, smoother.

[43:17] So you can see that there's less fluctuation, there is less variability in the, in the resulting, um, uh, plot, which makes it easier to see the patterns.

[43:35] Uh, we might be interested to upsample our, uh, time series data as well.

[43:43] That is when we don't have enough samples, when we don't have, um, enough data points.

[43:49] So we would want to increase the number of data points for different reasons.

[43:54] For, for example, the purpose of training, uh, training a, um, a machine learning model.

[44:00] Um, sometimes, um, some machine learning models perform better with, uh, more data points or more samples.

[44:11] So we might want to up-upsample our, our time series data.

[44:16] Uh, that involves increasing the frequency of the time series data, as we explained before.

[44:21] An example of that is converting daily data to hourly data, for example, or, um, yeah, s-things like that.

[44:31] And then we can-- uh, that can be done using, uh, techniques such as inter, uh, interpolation and, uh, and, uh, we can also use, um, uh, filling missing values.

[44:45] Uh, we can use techniques such as, for example, forward filling to increase the number of the data points or increase the frequency of the, uh, uh, time series data.

[45:00] So, um, we can apply the up-sampling using the resample function again.

[45:08] Uh, you can see that, um, here we call, uh, call the resample function.

[45:14] We pass, uh, the specified rules to be twelve, twelve hours, which means, uh, for every day, um, in the original data frame, we get two new data points, um, in, in the, in the resulting data frame, which correspond to, um,

[45:29] the first and the s- the first and the second twelve hours of the first twelve hour and the second twelve hours of that date.

[45:37] Uh, which means we are going to have more data points, but we do not have values for all those data points.

[45:44] We only have values for one of the data points of, uh, for, for every date, uh, that, that, that is either the first twelve hours or the second twelve hours.

[45:54] So the second, um-- the way the data frame is created is that the, the value, the, the acres burned value of the second, um, uh, second, uh, twelve hours is set to NaN or not a number.

[46:12] Uh, and once the function sum is applied, uh, um, that value is, uh, uh, is changed to zero.

[46:20] So, uh, this zero here specifies that, uh, we, we haven't had any values here.

[46:28] Uh, we haven't had any acres burned value for the second twelve hour of, uh, this date.

[46:34] Um, you can, you can see the same thing here.

[46:40] So again, uh, when you upsample a time series dataset, new rows are inserted into the data frame for the missing time periods.

[46:49] Uh, since the original dataset has a lower frequency than the resampled data, some of the new rows will have, uh, missing data, uh, which will be filled with NaN, as I mentioned.

[47:03] In this case, the sum aggregation method is used to calculate the sum of the acres burned values for each twelve hours period.

[47:11] And, uh, when the sum, sum function en- uh, encounters a NaN value, it treats it as a zero, uh, which can lead to, uh, some zero values in the resampled data frame, as you can see here.

[47:26] Okay.

[47:27] So how to fix a missing values issue that's, uh, caused by upsampling.

[47:34] So the problem was that we didn't have value for the second twelve hours of a day.

[47:39] Um, to fix that issue, we use, uh, we can use, uh, forward filling.

[47:45] So we call the forward fill function after we're s- we resampled.

[47:51] In this case, we know we are doing upsampling, uh, because we are increasing the frequency of the data.

[47:56] So, um, the value that's, uh, the value of acres burned that's been specified for the first twelve hours will be forward-filled to the second twelve hours, and they will have values for all the, uh, data points in the, in the

[48:14] data frame.

[48:15] As a result, when we apply the sum function here, uh, we get the, uh, we get the exact same value, uh, that we had for the first, for the first twelve hours of a day, for, for the second twelve hours

[48:29] as well, uh, or second twelve hours as well.

[48:32] And, uh, this will happen for the next day as, um, in, in, in, in the exact similar way.

[48:39] So, uh, for this-- for, for the second day of January We have one to two point five acres burnt and we get in, in the first twelve hours, uh, and we have the same number for the second twelve hours as

[48:56] well.

[48:56] So we four field the value, uh, in each day from the first twelve hours to the second twelve hours.

[49:04] And, uh, yeah.

[49:08] Okay, so the next topic we're going to talk about is, uh, rolling windows.

[49:17] Rolling windows are to compute, uh, aggregate statistics over a moving window of time, where-- or in other words, over a moving interval.

[49:27] Right?

[49:29] So you can move this interval over the, over the time, uh, or, uh, x-axis and, uh, and apply some, uh, aggregate statistics to each period.

[49:44] That involves, uh, sliding a window of fixed size over the time series data and applying function to the data within each window.

[49:52] It's useful for smoothing noisy data and detecting ano-uh, anomalies.

[49:57] Uh, an example is a rolling window can be used to compute a moving average or moving mean of a time series data set.

[50:08] Uh, the moving average at each time point, if you want to calculate the moving average, the moving average at each time point is computed by, uh, taking the average of the data within, uh, within the interval, within the fixed, uh,

[50:25] window size that we specify.

[50:27] And of course, we are sliding that interval, uh, uh, alongside, uh, or along the time axis, which is x-axis in this, in this graph.

[50:44] Okay.

[50:47] So here's an example.

[50:48] Uh, so first we have, um, here we are not doing any, any rolling.

[50:52] This is our, um, original data frame, um, that captures these two values, high and low, for stock data, uh, for dates that are before this particular date, uh, which is the end of January 2020.

[51:15] And so this is the visualization of the original data or the original time series data.

[51:25] And here we do-- we apply the rolling.

[51:29] So as you can see, uh, again, the same query is executed, but rolling function is applied.

[51:36] So .rolling window equals seven.

[51:40] That means the width of the window will be seven days.

[51:45] Okay?

[51:46] And mean periods equals seven means we need to have at least seven, uh, data points to, um, to apply the, the aggregate function mean.

[52:00] So that's the minimum requirement.

[52:01] That's the minimum number of, uh, uh, data points that we need to have to calculate the mean.

[52:08] So we're setting window to seven, also mean periods to seven.

[52:12] Uh, and then the next step, of course, is visualization.

[52:17] So we're using from Seaborn the replot function and, uh, the frequency of...

[52:25] Uh, so here we are also creating a date range to specify the frequency of the data that is going to be displayed.

[52:31] We're going to only display the data by, uh, business days.

[52:35] So we only show the business day-- the data that's been collected during the business days.

[52:40] Okay?

[52:41] And yep.

[52:45] So we, we apply that to the x ticks, so we can, um, so we know what is going to be displayed there, the business days.

[52:53] And the labels, of course, correspond to the same, uh, business days.

[52:56] Uh, we also need to, uh, format the business days in, in, in whatever format we prefer.

[53:03] In this case, we prefer, uh, this format.

[53:06] So we format them as month, uh, uh, dash e, uh, month dash day.

[53:14] Uh, you can see here we have month and we have day.

[53:17] So this is sixth of January, seventh of January, eighth of January, and so on.

[53:22] So what's happening here is that the average of, uh, values, for example, for the-- for this column, for low, the average of these seven values will be computed.

[53:35] And here you get the average value for low.

[53:39] Similarly, the average value for high will be calculated for these seven days because the size of the window is seven, right?

[53:46] So for these seven days, the average value will be calculated and will be placed here for high as well as for low.

[53:55] So you get the average low and the average high for those seven days.

[53:59] One, two, three, four, blah, blah.

[54:01] So the seven days here that's been, that's been specified here.

[54:05] So here you get the seventh entry here, uh, includes the average value of high before, uh, before and including this data point and the average value of low before including this data point.

[54:21] Okay?

[54:23] And this will continue.

[54:24] Uh, why are we getting NaN values here?

[54:27] Because obviously for the first row, we don't have, we don't have, um, enough number of data points to calculate the mean, and mean periods is seven, right?

[54:38] So how many rows are coming before this row?

[54:41] Zero.

[54:43] We only have one row here, so we don't calculate the mean.

[54:45] That's, that's why we're getting NaN for high and low, a-and low, right?

[54:49] The next data point, still we, we only have one row before that.

[54:53] Uh, so we all together we get two rows, which is not at least seven.

[54:57] So we get NaN again for high and low, and so on and so forth.

[55:02] So the f-- uh, here's the first row or f-first data points where we can actually calculate the mean because there are enough data points available to calculate the mean.

[55:11] So there are six here, and this is the seventh, and we calculate the mean.

[55:18] Okay?

[55:20] So this is what we mean by window seven and mean period seven.

[55:26] And as, as you can see, the, and the resulting graph would be something like this.

[55:31] Uh, this is the previous graph or the previous plot based on the original data set that included the, the actual values for high and low.

[55:39] But here we are showing the, the average high and the average low, um, for, for the time period or for the time window that is specified.

[55:57] Okay, so here's an example of using rolling windows in real world.

[56:02] You can see this data again is extracted from, uh, from gov.au, health or healthgov.au.

[56:10] This is, uh...

[56:11] this shows COVID-19 associated death is, uh, rolling seven-day average by, uh, jurisdiction from 1st of January, twenty twenty-two to twenty-fifth of April, twenty twenty-three.

[56:24] So you can see it by different states.

[56:26] You can see the number of, uh, the number of deaths, for example, in Victoria, you can see it in this color, and the number of deaths in New South Wales.

[56:35] So you can see that dif-different colors are used for different states and, uh, territories.

[56:41] And, uh, you can see that over the period of, you know, from, for the period that's specified here, and this is the rolling seven-day average.

[56:50] So the, the, what is the width of the window here?

[56:52] What is the size of the window?

[56:54] Seven days.

[56:55] Okay?

[56:55] So it's the seven days average that has been calculated for the number of deaths.

[57:00] Uh, if you go to the link that I have provided here, you can see lots of interesting, uh, visualization or plotting of COVID-19 related data from, uh, uh, from, uh, from health.gov.au, and you can try to map them to what

[57:14] we, uh, we're teaching today.

[57:19] Here's another example.

[57:20] You can see that, uh, uh, I'll let you to spend time on that yourself.

[57:31] Expanding and running totals.

[57:34] So now we talk about expanding and, uh, running totals is a special case of expanding.

[57:47] So what is the concept of expanding?

[57:50] Expanding is a method for calculating, uh, um, a, a cumulative metric in, uh, time series data.

[57:59] So here's the difference between expanding and rolling.

[58:03] Unlike rolling or unlike a rolling window, which calculates the metric over a fixed time window, the expanding method calculates the metric over all, uh, preceding time periods.

[58:15] So up to that point, not a, not a fixed period, not a fixed time period, not, not the last seven days, for example, but up to that day, right?

[58:26] Uh, you start with the value of the metric at the first time period and then calculate the cumulative value of the metric for all the, uh, preceding time periods.

[58:40] Uh, the example is, um, the expanding sum of the time series, uh, would be the sum of all values up to and including the current time period.

[58:51] And the expanding method is useful for analyzing long-term trends in, uh, time series data and, uh, useful also for identifying patterns that may not be visible in shorter time, time windows.

[59:04] It can also help to smooth out short-term, uh, fluctuations and highlight long-term or longer-term trends in the data.

[59:17] So running totals, uh, are, are type of expanding calculations that involves adding up the values of the time, time series over time and, uh, in other words, the sum of the values, and that starts from the first time period and

[59:35] includes all the previous values up to the current time period.

[59:41] Running totals are often used to track cumulative changes or trends in the data and to identify key, um, inflection points or anomalies in the, uh, anomalies in the time series data.

[00:04] Okay, so here's an example.

[00:06] We have, um, we have used...

[00:09] this time we use expanding, so instead of rolling, we use expanding.

[00:14] Again, we have data frame that we have, uh, created from, you know, this people file, and then we apply expanding function to it.

[00:22] So acres burned.expanding.sum.

[00:25] It's going to sum up the values, uh, for each d- for each data point, it's going to sum up the values up to that point.

[00:32] Uh, so you can see what's happening here by looking at the data.

[00:36] You can see that, um, um, twelve eighty point seven six is added to one two two point five zero, um, and this forms the new value one four zero three point two six.

[00:51] So this is the summation of...

[00:53] Let me get a pen.

[00:55] So this value here is the summation of these two values.

[01:00] You can see they are highlighted in green.

[01:03] And then this value here is summation of these four values.

[01:08] So again, from the beginning.

[01:11] So it adds up, in other words, summation of this and this.

[01:17] Right?

[01:18] So we sum up the values from the beginning up to that point to find the new value.

[01:26] So this value is the summation of all the previous points and the current point, and the, and the, and the value from the current point, which is, uh, listed here.

[01:38] And this value is the summation of this value and all the points before that, which is this one, or all the values from the points before that, which is this one, and then the value is this.

[01:48] So how do we calculate this value here?

[01:52] We just add up all these numbers, this one, this one, this one and this one, and it gives us the new value.

[02:02] Right?

[02:04] So, uh, it's a cumulative sum of the previous values, uh, that, uh, is captured, this new, new, new column, running totals.

[02:15] And this column has been created here.

[02:18] Acres burned running totals equals acres burned at expanding.

[02:23] So this new sum will be, will be stored in this newly created column, uh, running totals.

[02:30] All right?

[02:31] So this is an example of s-expanding and, uh, we, we call this, uh, running totals, uh, which is a special type of expanding.

[02:40] But instead of sum, we might have applied, for example, average, mean, mean function, right?

[02:47] So it depends, uh, or, or a custom function, right?

[02:50] We could have applied a custom function.

[02:53] So we can actually apply functions to expanding.

[02:57] Um.

[03:02] All right.

[03:08] How to plot running totals.

[03:10] So plotting running totals, uh, is, uh, is straightforward.

[03:15] We can just-- in this example, you can see the structure of the data.

[03:20] This, the data is stored like this.

[03:22] We have running totals that we just calculated, and running totals are calculated based on acres burned, obviously.

[03:30] Um, so if you want to plot this, obviously we, we want the discovery date still in the, in the plot.

[03:37] Uh, we're assuming that we need to, to show the dates on the, on the plot.

[03:41] This is, this, this is, uh, we, we are visualizing time series data, so time or dates in this case need to be there in the, in the plot.

[03:50] So we need discovery dates, and we also need acres burned and running totals and the corresponding values.

[03:57] So one easy way that we can do that is that we can melt the data frame.

[04:02] So we can melt this data frame into the, into the, into this one by...

[04:07] We have seen examples of this in the previous lecture.

[04:10] So pandas.melt and then pass the data frame that is, um, in this case acres burned, this data frame.

[04:19] And we specify the id vars, the identifiers.

[04:23] The identifier in this case is, uh, discovery date.

[04:27] So this discovery date will be as used as the identifier.

[04:30] And then we also specify the value vars.

[04:32] These are the two value vars, uh, acres burned and acres total, and, and sorry, running total.

[04:38] Running total.

[04:39] So running total, acres burned.

[04:41] These are the, uh, the, the value vars.

[04:44] And, uh, we, we, we use this, uh, value type for the name of the new variable that will be created.

[04:52] So we have, uh, we have value type here.

[04:57] So value type will take either of these two values, acres burned or running total.

[05:03] Right?

[05:04] So in this case, you can see the acres burned here, acres burned, acres burned, acres burned.

[05:08] And you can have some, uh, running totals, running totals, acres burned.

[05:12] So these are the values that value type can take.

[05:16] So value type is either acres burned or running totals.

[05:20] And the values that correspond to those, um, um, th-those values, that means th-those value types, uh, are of course, um, uh, uh, the numbers that have, that have been extracted from this column.

[05:35] So, um, so this is melting.

[05:38] If you're not sure how this works, please refer to the previous lecture and I have, uh, the, um, I have explained it in the, in the recording.

[05:47] Um, so we convert this data frame to this data frame.

[05:53] In other words, we melt the data frame into this format, which makes, makes it easier to visualize.

[05:59] Uh, and here's the visualization part.

[06:03] We also showed this in the previous lecture.

[06:05] We use the cat plot to A plot categorical data.

[06:09] This is used for Seaborn.

[06:11] And, uh, the kind is a kind of the plot that we're using here is a bar plot and, uh, d- and what do you call it?

[06:20] Uh, the x-axis is going to show the discovery date, right?

[06:24] Uh, so the discovery, remember we used it as the identifier to melt the data frame, and we still have it here in the resulting data frame, and the acres melted.

[06:34] So we use discovery date, uh, as the, uh, on our x-axis.

[06:39] And then we have values, which are the values that correspond to either acres burned or running totals, the values that correspond to the value types.

[06:50] Remember, value type can be either acres burned or running total and of course, each of these have their own values.

[06:59] And you can see that we also use, um, uh-- Yeah, so, so we specify this and then, um, the bar plot, the resulting bar plot is showing, um, is distinguishing between different, different value types by different colors.

[07:15] So we have two bars.

[07:17] Uh, the orange bar corresponds to running total, and the blue bar corresponds to acres burned.

[07:24] So you can see that the acres burned are, um, they, they have different values for different discovery dates.

[07:31] So here we have a value, and here we have a smaller value, and we have a larger value.

[07:36] So there, there are different values here.

[07:38] But you can see that the running total is always increasing, okay?

[07:44] Because it's a cumulative value.

[07:46] It's the summation of...

[07:48] So in other words, this, um, for example, let's say this one is calculated by summing up this blue bar here, this one here, and this one here.

[08:05] Right?

[08:06] So this one's the summation of all these three.

[08:11] Okay?

[08:13] Uh, obviously for the first one, they're the same.

[08:16] And the second one, so this one, let me choose a different color.

[08:21] This one is calculated by summing up this value with this small value here.

[08:31] So you can see this bar is slightly higher than this one, okay?

[08:37] And so on and so forth.

[08:38] So you can see overall, overall, the value of running total is increasing.

[08:52] But obviously, the values a- acres burned can in- increase and decrease, increase, increase, decrease, stay around the same, decrease, decrease, increase.

[09:05] But the o-- the summation, the cumulative sum of acres burned is increasing over time.

[09:12] Okay.

[09:15] All right.

[09:16] So this was the example that show how we can plot our running totals.

[09:22] Remember, running totals is a special type of expanding, and we also explain what is the difference between expanding and rolling.

[09:31] Where in, in expanding, we, uh, accumulate, uh, the, the aggregation over, over the time period.

[09:40] Uh, start, start from, uh, we start from the beginning of the, uh, the per-- the, the beginning of the time or the, the range up to the current point, and, uh, we calculate the value for, uh, for, for the whole,

[09:56] for the whole interval, not for a fixing and not, not for a fixed size interval.

[10:03] Uh, whereas for rolling, we would, uh, we would slide the fixed time, fixed size, uh, interval, uh, along the x-axis and calculate the, uh, calculate the, uh, for example, the sum or the mean or whatever we are calculating for that

[10:21] time interval.

[10:23] So you can go back and, uh, and see the differences and try to do it yourself.

[10:28] We have provided examples.

[10:30] So you can go and, um, and, uh, what do you call it?

[10:36] Uh, open the Jupyter notebook file and execute the examples and see what happens.

[10:46] Uh, there's one more video that, uh, we have here that shows, um, the, uh, that, that summarizes, uh, some of the things that we discussed, especially about, uh, expanding and, uh, rolling windows.

[11:03] It also talks about shifting, uh, which we didn't cover in this lecture, but you can, you can understand the, the, the, um, topic by watching this video.

[11:14] So it's a good summary of what we discussed.

[11:17] So again, I recommend watching and, uh, there are some, uh, good examples there.

[11:27] All right, so this will, uh, wrap up our today's lecture.

[11:31] I'm going to, uh, stop at this point and, uh, and see you next time.